MLNS - Report

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Learning to Identify Graph Generative Mechanisms: A Comparative Study of Machine Learning Approaches for Graph Classification

1 Motivation and Problem Definition

Networks arising in social, biological, technological, and informational systems often exhibit structural patterns that reflect the mechanisms by which they were formed. Classical models such as the Erdős–Rényi random graph, the Barabási–Albert preferential attachment model, the Watts–Strogatz small-world model, and the Stochastic Block Model capture distinct generative principles, including randomness, growth and preferential attachment, local clustering with short path lengths, and community structure.

The goal of this project is to investigate whether machine learning models can accurately infer the generative mechanism of a graph based solely on its structural properties. Formally, given a graph G=(V,E) generated by one of several classical models, we aim to learn a classifier that predicts the underlying generative model.

Understanding generative mechanisms is central to network science. Many real-world networks display properties such as heavy-tailed degree distributions, clustering, small diameters, or modular structure. Inferring which generative principles best explain observed networks has implications for Network Robustness Analysis for instance.

Moreover, this project provides a controlled experimental framework for understanding what structural features different learning approaches truly exploit.

2 Methodology

To analyze generative processes, we will create a synthetic dataset of generated graphs from four classical models (Erdős–Rényi (ER), Barabási–Albert (BA), Watts–Strogatz (WS), Stochastic Block Model (SBM)).

To avoid trivial classification based on density, we want all the generated graphs to have approximately the same density, regardless of the model used for generation.

We will compare two families of methods :

- Structural features (for an exhaustive list of all features, see the custom Python module `generative_functions.py`)+ Classification ML algorithms (such as SVM and Random Forests (RF)). This approach tests whether global statistics suffice.
- Graph Convolutional Networks (GCN) with global pooling to obtain graph-level representations (for more details about the GCN, see the custom Python module `gcn_functions.py`.

3 Evaluation

The dataset will be synthetically generated using Python (NetworkX), this allows full control over parameters and reproducibility. We will generate approximately 200 graphs of size $n = 200$ per model and then separate this dataset into train, validation and test splits, stratified by the type of graph.

To evaluate each approach, we will use the Classification accuracy, the Macro-F1 score (a multi-labels prediction/classification metric), as well as the Confusion matrix.

Concretely, the project will contain the following experiments :

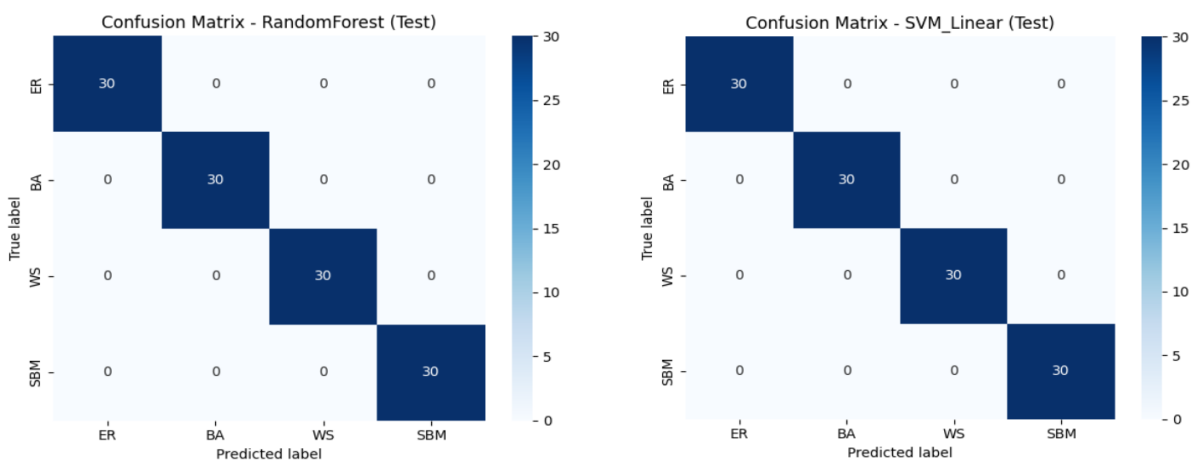
- A baseline experiment, without constraints on the generative models parameters to ensure a similar density. This should lead to a easier classification but a poor generalization of the classification models if the densities of the graphs happen to be similar.
- A density-controlled experiment, to ensure the densities of graphs generated from different algorithms have a similar density and that the classification models will leverage characteristics beyond the average degree (for instance spectral features, degree distribution, global structure).
- A density-controlled experiment + training and test on graphs of different sizes ($n = 100$ for training and $n = 200$ for test). This allows to evaluate if the models are robust for graph of different (and unseen) sizes.
- A density-controlled experiment + random edge perturbations to see if the models are robust to more realistic graphs (with errors and noise in the connections).

4 Experiments

4.1 Baseline Experiment

In this experiment, we set the average degree of each type of graphs to different value (4 for ER graphs, 7 for BA graphs, 10 for WS graphs, 12 for SMB graphs).

Both the Random Forest algorithm and the SVM algorithm reach a perfect accuracy and F1 score, which could have been expected given that each graph type has a distinct average degree value.



(a) (b)

Figure 1: Baseline Experiment - RF & SVM - Confusion Matrices

However, when we dig a bit deeper, it appears that the `mean_degree` feature is not the most informative one for the algorithms. Let's see that for the RF algorithm:

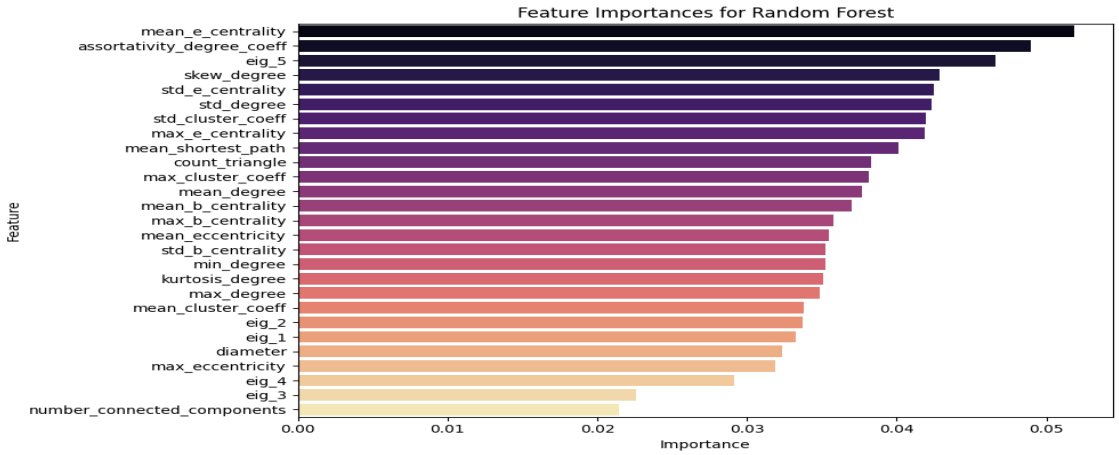


Figure 2: Baseline Experiment - RF - Features Importance

The features `mean_e_centrality`, `assortativity_degree_coeff` and `eig_5` are the most informative one. We thought that the feature `mean_degree` would be really impactful since the mean degree of each type of graph was set to a different value. Actually, the importance of each feature is provided relatively to the other's, so `mean_degree` might be very impactful, yet less than `mean_e_centrality` or `assortativity_degree_coeff`.

The GCN also reaches a perfect accuracy and F1 score :

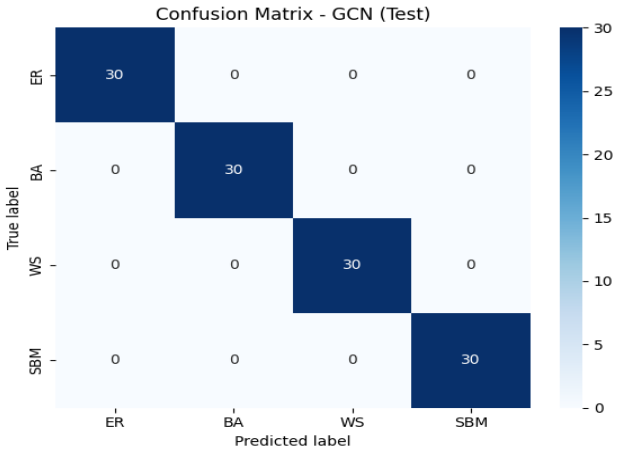


Figure 3: Baseline Experiment - GCN - Confusion Matrix

When we analyze the importance of each feature of the GCN (by gradient), the one that contributes the most to the class prediction (almost 2 times more) is the betweenness centrality (so how frequently a node appears in the shortest path between two other nodes) :

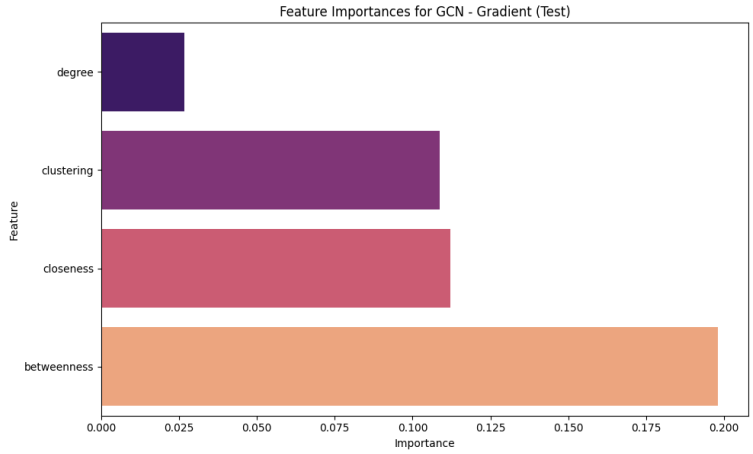


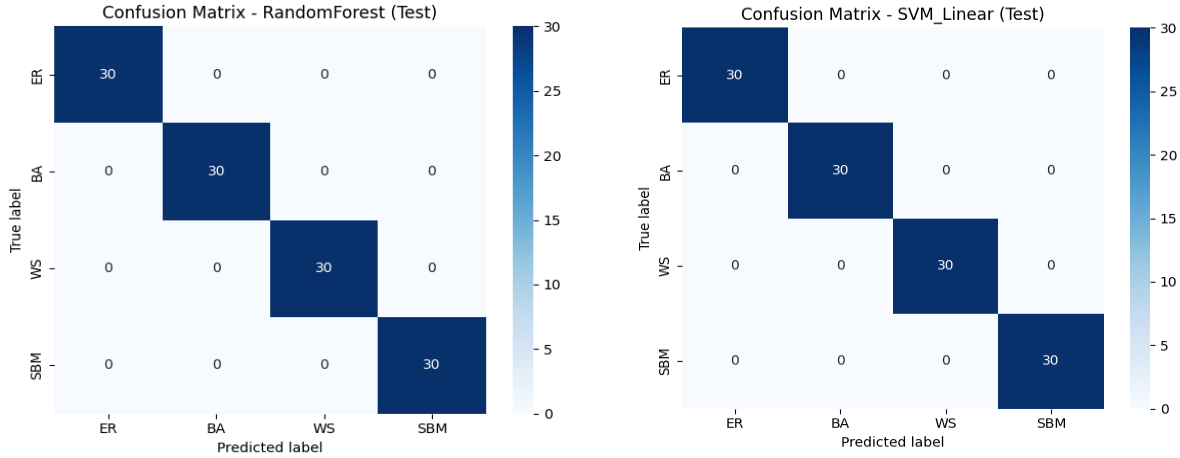
Figure 4: Baseline Experiment - GCN - Features Importance

The GCN directly uses structural features that are more complex that just the degree of each node. Indeed, the degree of each node + the average pooling could have indicated that, for each type of graph, the average degree was different.

4.2 Density-Controlled Experiment

In this experiment, we set the average degree of each type of graphs to the same value : 8.

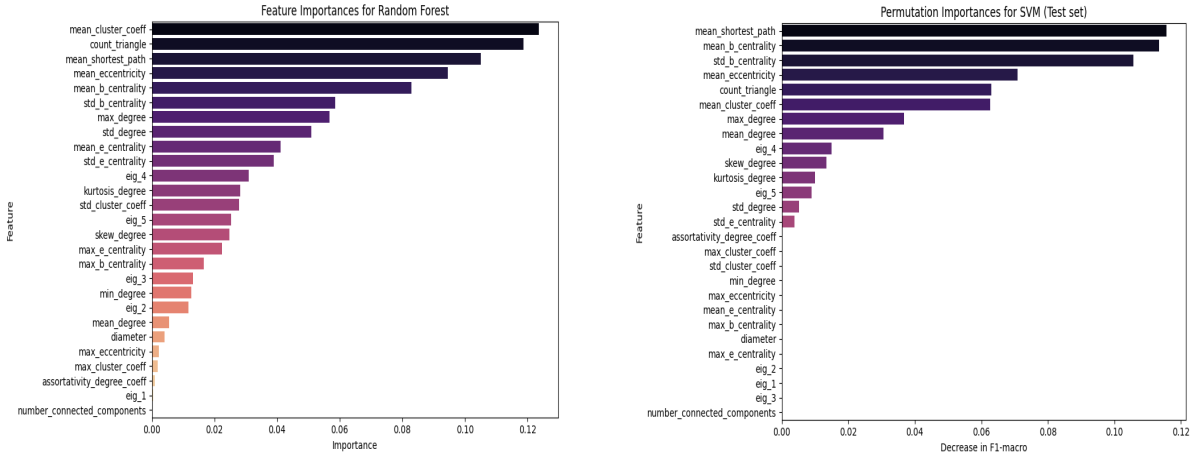
As in the first experiment, both the Random Forest algorithm and the SVM algorithm reach a perfect accuracy and F1 score.



(a) (b)

Figure 5: Density-controlled Experiment - RF & SVM - Confusion Matrices

As expected, for both algorithms, the `mean_degree` is far from being the most informative feature (since all graphs have the same average degree). The RF model relies more on `mean_cluster_coeff`, `count_triangle` and `mean_shortest_path`, so more on how structured and how far the nodes are. Same observation for the SVM model since it relies more on `mean_shortest_path`, `mean_betweenness_centrality` and `std_betweeness_centrality`.



(a) (Features Importance) (b) (Features Permutation)

Figure 6: Density-controlled Experiment - RF & SVM - Features Importance

The GCN also reaches a perfect accuracy and F1 score :

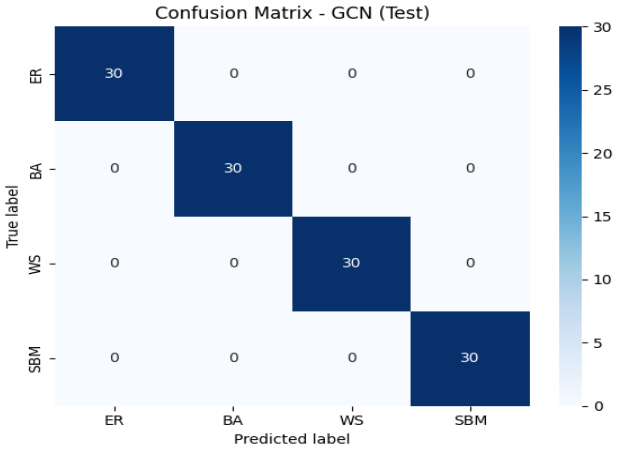


Figure 7: Density-controlled Experiment - GCN - Confusion Matrix

However, where the RF and SVM models leverage global metrics of the graphs, the GCN manages to perfectly classify graphs by only leveraging local information (even if it summarizes this local information through mean-pooling), especially the `clustering_coefficient` of each node. We can note that `clustering_coefficient` only gives information about the structure of the graph (and not about the distance between the nodes).

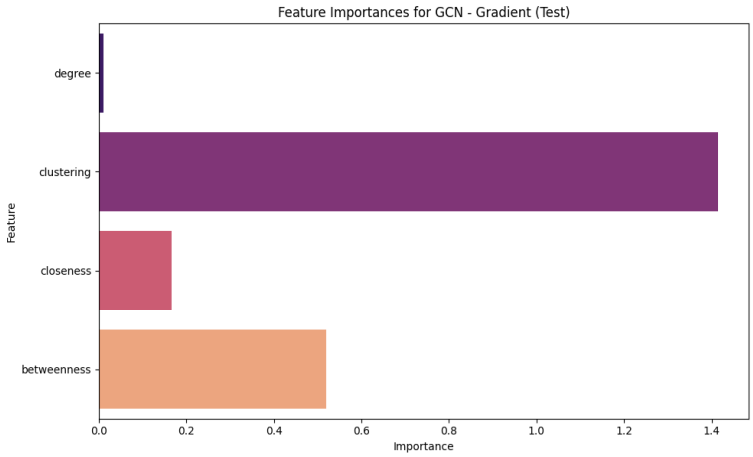
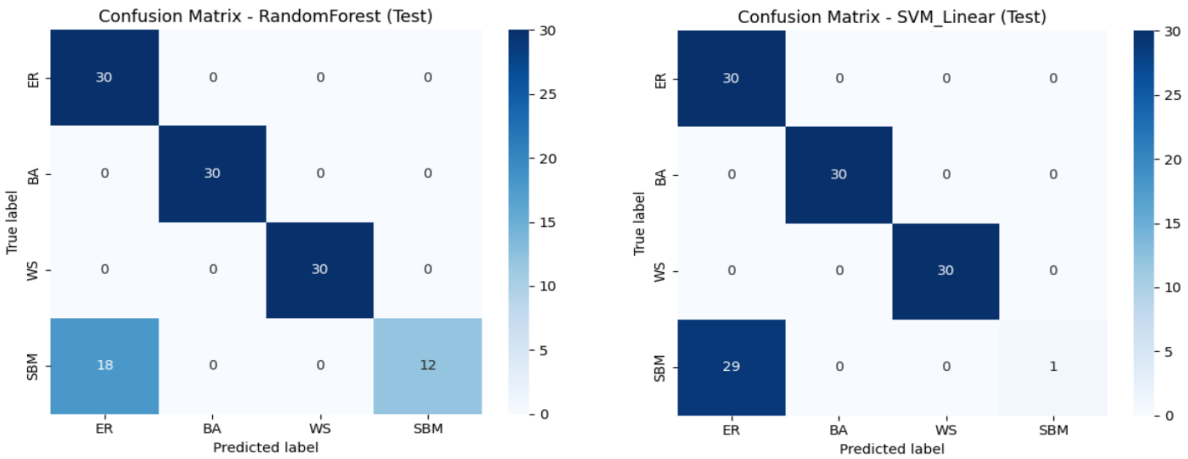


Figure 8: Density-controlled Experiment - GCN - Features Importance

4.3 Density-Controlled Experiment + different sizes of training & testing sets

In this experiment, we set the average degree of each type of graphs to the same value : 8. Moreover, the size of graphs in the training set is set to $n = 100$ and the size of graphs in the test set is set to $n = 200$. In this experiment, we want to see if the model can generalize on graphs of different size.

However, it appears that all models correctly classify ER, BA and WS graphs but struggle to correctly classify SBM graphs. Only the RF model manages to classify roughly half of the SBM graphs correctly when the SVM model and the GCN just cannot do that.



(a) (b)

Figure 9: Density-controlled + different test size (train = 100, test = 200) Experiment - RF & SVM - Confusion Matrices

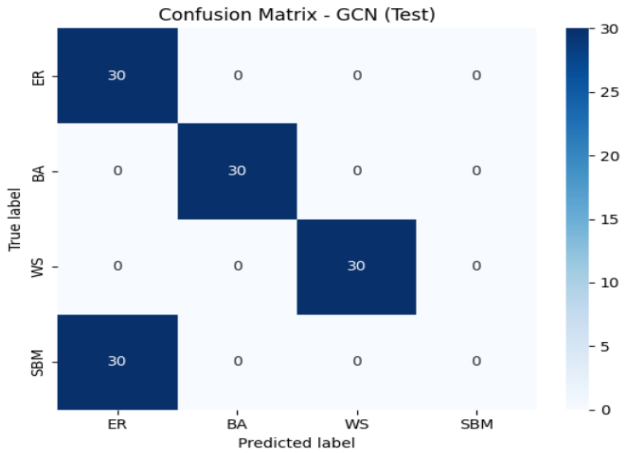
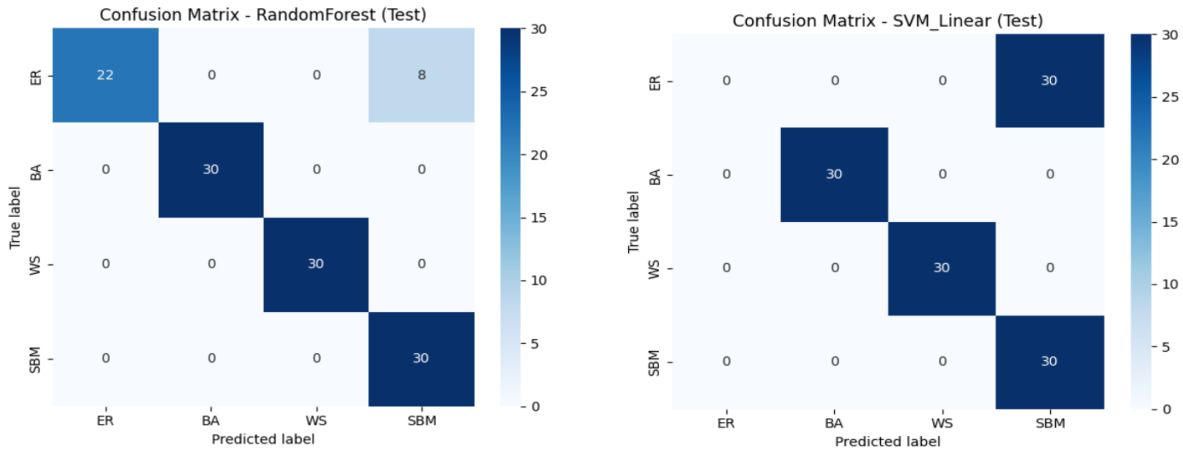


Figure 10: Density-controlled + different test size (train = 100, test = 200) Experiment - GCN - Confusion Matrix

The result is similar if we invert the train split and test split sizes (training on graphs of size $n = 200$ and evaluation on graphs of size $n = 100$), even if the RF model is actually performing well.



(a) (b)

Figure 11: Density-controlled + different test size (train = 200, test = 100) Experiment - RF & SVM - Confusion Matrices

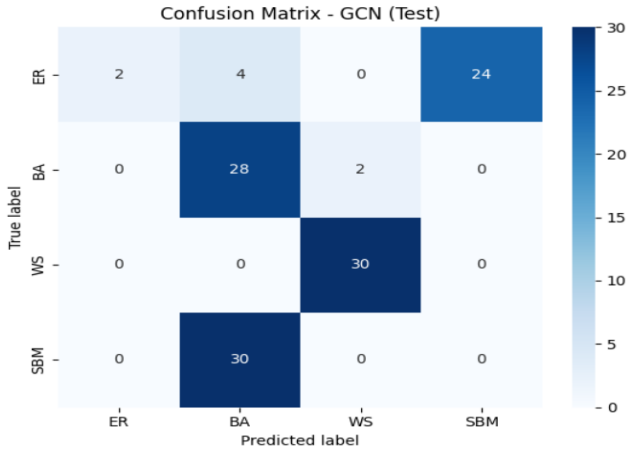


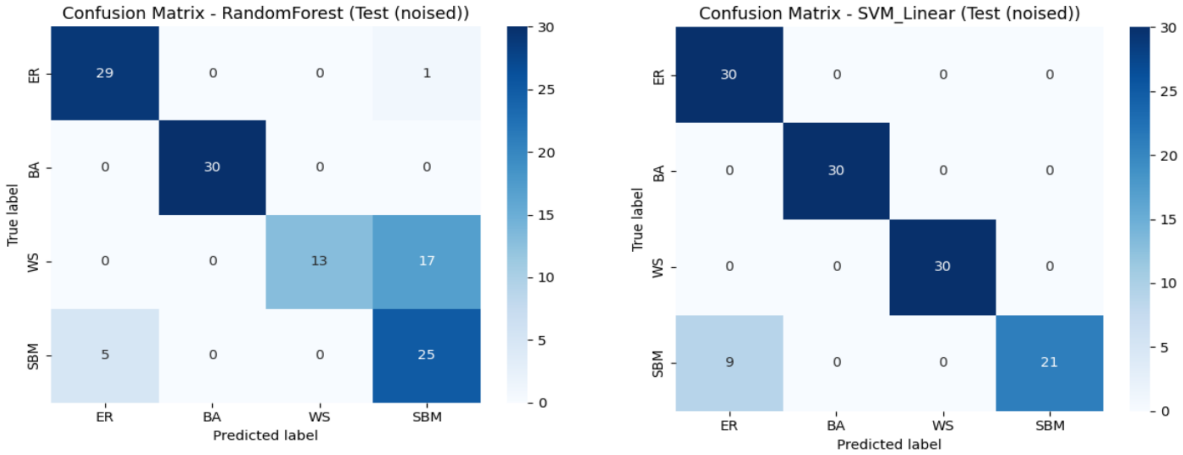
Figure 12: Density-controlled + different test size (train = 200, test = 100) Experiment - GCN - Confusion Matrix

As we can see, this lack of robustness is not related with a specific type of graph (struggle to classify SBM graphs for train = 100 & test = 200; struggle to classify ER graphs for train = 200 and test = 100). It seems that the models struggle to correctly classify graphs of different sizes because the weights (of each feature) they were computing during the training phase are not relevant anymore for a graph with a different size : the features that should be used are not the same if the size of the graph changes.

5 Density-Controlled + Random Edge Perturbations Experiment

In this experiment, we set the average degree of each type of graphs to the same value : 8. Moreover, we set a edge perturbation probability at 10%, which means that 10% of the edges will be deleted and roughly the same amount of edges will be randomly added to the graph.

Each model was training on clean graphs and tested on both clean and noised graphs.



(a) (b)

Figure 13: Density-controlled + Random Permutation (10%) Experiment - RF & SVM - Confusion Matrices

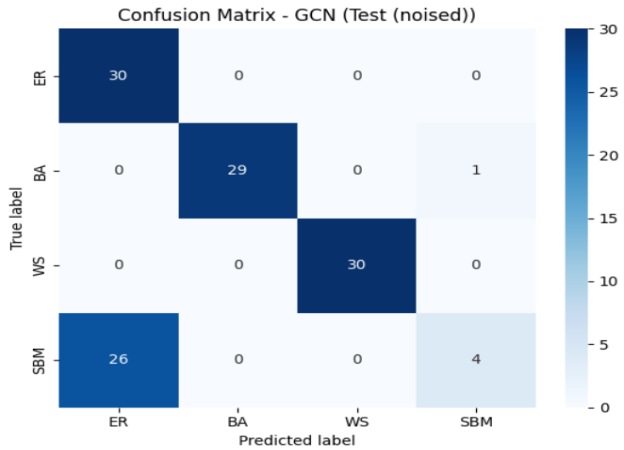


Figure 14: Density-controlled + Random Permutation (10%) Experiment - GCN - Confusion Matrix

We see that all models struggle to classify SBM graphs. This can be explained by - and could be expected given - the nature of this type of graphs. Indeed, the SBM graphs (in our case) are structured around two communities that have many connections between their nodes but very few connections with the nodes of the other community. When we randomly select 10% of all the edges and rewire them between two random nodes, the new edge has as many chance to be linking two nodes of the same community as to be linking two nodes of different communities. This goes against the way a SBM graph is initially generated with a connection between two nodes being much more likely if those nodes belong to the same community. The random perturbation then heavily change the structure of the SBM graphs, which explains why the models struggle to classify this specific type of graph.

6 References

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